

Phishing URL Detection Using Machine Learning

Begimai Alisherova

Thompson Rivers University

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Piper Jackson

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1. Data Description

The dataset, titled PhiUSIIL Phishing URL Dataset (1), is designed to classify phishing URLs and legitimate websites. It was acquired from the UCI Machine Learning Repository and is structured in a tabular format. The dataset contains 235,795 records with 54 features, capturing a wide variety of URL and web page attributes. To download the dataset, a link is attached in the references (2).

The features include URL-based attributes such as URL length, top-level domains (TLDs), HTTPS usage, and obfuscation techniques. These attributes help detect patterns in phishing URLs, which often use uncommon TLDs, obfuscated characters, or IP addresses instead of domain names. The dataset also contains HTML-based features such as the presence of page titles, favicons, and descriptions, along with indicators of suspicious behavior, like excessive redirects, hidden fields, and external form submissions. Specific financial terms, such as “bank”, “pay”, and “crypto”, are also flagged to detect phishing websites targeting sensitive user information.

This dataset is complete, with no missing values, and was originally compiled for academic research in 2024, making it both reliable and recent. Its size and variety ensure it is highly applicable to machine learning tasks, particularly for building classification models to detect phishing URLs. Stored in Excel format, the dataset is ready for use with minimal preprocessing required.

2. Data Analysis

As mentioned before, the dataset has 235,795 records and 54 features. The first 5 rows of the dataset are printed below.

	FILENAME	URL	URLLength	Domain	DomainLength	IsDomainIP	TLD	URLSimilarityIndex	CharContinuationRate	TLI
0	521848.txt	https://www.southbankmosaics.com	31	www.southbankmosaics.com	24	0	com	100.0	1.000000	
1	31372.txt	https://www.uni-mainz.de	23	www.uni-mainz.de	16	0	de	100.0	0.666667	
2	597387.txt	https://www.voicefmradio.co.uk	29	www.voicefmradio.co.uk	22	0	uk	100.0	0.866667	
3	554095.txt	https://www.sfnmjjournal.com	26	www.sfnmjjournal.com	19	0	com	100.0	1.000000	
4	151578.txt	https://www.rewildingargentina.org	33	www.rewildingargentina.org	26	0	org	100.0	1.000000	

5 rows × 56 columns

Out of 54 features, 5 contain textual information. We have deleted these columns as they do not hold any significant information for us as all information was derived into new features. For example, “URL” of an object type was used for creating “URLLength”, “NoOfLettersInURL”, “LetterRatioInURL”, “NoOfDegitsInURL”, etc. Talking about the data types, the rest of the columns are float64 or int64 type.

```

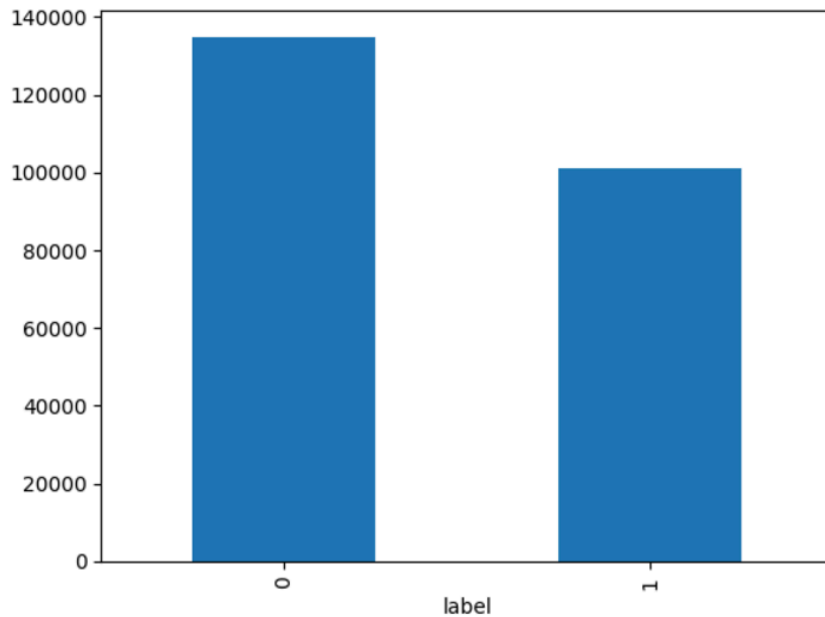
RangeIndex: 235795 entries, 0 to 235794
Data columns (total 56 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   FILENAME                             235795 non-null object
1   URL                                  235795 non-null object
2   URLLength                            235795 non-null int64
3   Domain                               235795 non-null object
4   DomainLength                         235795 non-null int64
5   IsDomainIP                           235795 non-null int64
6   TLD                                  235795 non-null object
7   URLSimilarityIndex                   235795 non-null float64
8   CharContinuationRate                 235795 non-null float64
9   TLDLegitimateProb                    235795 non-null float64
10  URLCharProb                           235795 non-null float64
11  TLDLength                             235795 non-null int64
12  NoOfSubDomain                        235795 non-null int64
13  HasObfuscation                       235795 non-null int64
14  NoOfObfuscatedChar                   235795 non-null int64
15  ObfuscationRatio                     235795 non-null float64
16  NoOfLettersInURL                     235795 non-null int64
17  LetterRatioInURL                     235795 non-null float64
18  NoOfDegitsInURL                      235795 non-null int64
19  DegitRatioInURL                      235795 non-null float64
20  NoOfEqualsInURL                      235795 non-null int64
21  NoOfQMarkInURL                      235795 non-null int64
22  NoOfAmpersandInURL                  235795 non-null int64
23  NoOfOtherSpecialCharsInURL           235795 non-null int64
24  SpacialCharRatioInURL                235795 non-null float64
25  IsHTTPS                              235795 non-null int64
26  LineOfCode                           235795 non-null int64
27  LargestLineLength                    235795 non-null int64

```

Additionally, we looked into the distribution of labels to see if the dataset is imbalanced.

It turned out they used 1 to label legitimate websites and 0 for phishing websites. We have

changed the labels, so now we have 1 for phishing and 0 for legitimate websites. Around 135k records are legitimate and around 100k websites are phishing. From the bar chart, we can see that the dataset is fairly balanced and both classes are well-represented.



From the summary statistics (see the image below), we see that the dataset has significant variance across its features as each column has a distinct range and distribution. Numerical features like URLLength and URLSimilarityIndex have wide ranges and high variability, while binary features such as IsDomainIP and HasCopyrightInfo are more consistent. The differences in how the features are distributed indicate the diversity of the data. The dataset's diversity makes it suitable for phishing detection. Such variability means that we will have to scale our data to ensure fair representation for Principal Component Analysis (PCA).

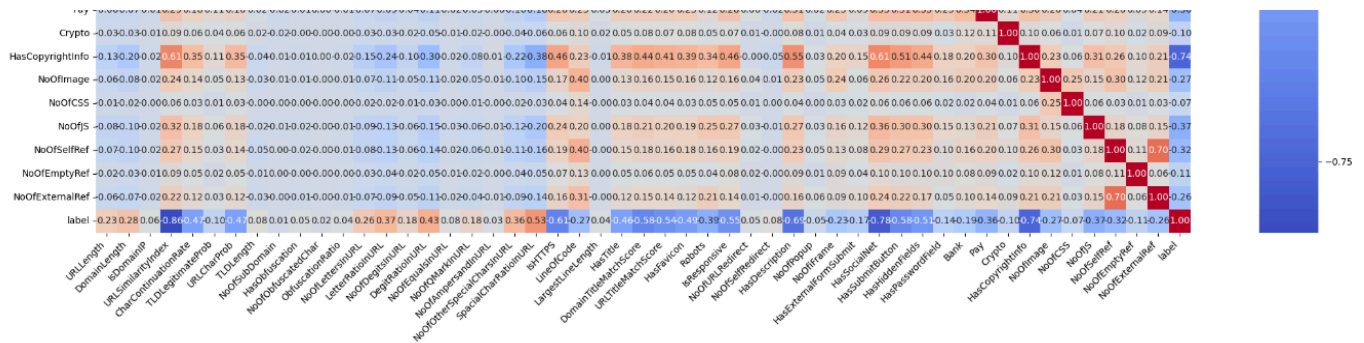
	URLLength	DomainLength	IsDomainIP	URLSimilarityIndex	CharContinuationRate	TLDLegitimateProb	URLCharProb	TLDLength	NoOfSubDomain
count	235795.000000	235795.000000	235795.000000	235795.000000	235795.000000	235795.000000	235795.000000	235795.000000	235795.000000
mean	34.573095	21.470396	0.002706	78.430778	0.845508	0.260423	0.055747	2.764456	1.164758
std	41.314153	9.150793	0.051946	28.976055	0.216632	0.251628	0.010587	0.599739	0.600969
min	13.000000	4.000000	0.000000	0.155574	0.000000	0.000000	0.001083	2.000000	0.000000
25%	23.000000	16.000000	0.000000	57.024793	0.680000	0.005977	0.050747	2.000000	1.000000
50%	27.000000	20.000000	0.000000	100.000000	1.000000	0.079963	0.057970	3.000000	1.000000
75%	34.000000	24.000000	0.000000	100.000000	1.000000	0.522907	0.062875	3.000000	1.000000
max	6097.000000	110.000000	1.000000	100.000000	1.000000	0.522907	0.090824	13.000000	10.000000

8 rows × 51 columns

As for the data preprocessing steps, we have checked and found no missing values.

Outliers in this project are crucial as they may represent phishing attempts. Phishing URLs are often designed to deviate from normal patterns (e.g., very long URLs, excessive special characters), so we will keep all of them.

For checking the relationships between our features and target column, we have created a heatmap (see below). Using the correlation matrix, we can see which features are highly correlated with each other. From the bottom row that shows us correlations between the label column and every other column in our dataset, we can see that “label” is strongly correlated with “URLSimilarityIndex” (0.86), “HasSocialNet” (0.78), and “HasCopyrightInfo” (0.74). These features are significant in predicting whether the website is phishing or legit.

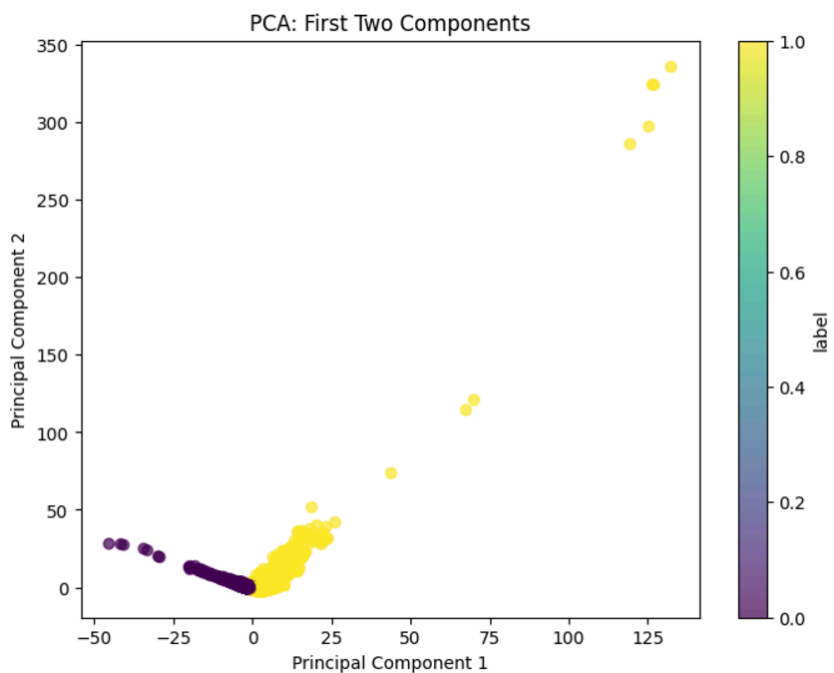


3. Data Exploration

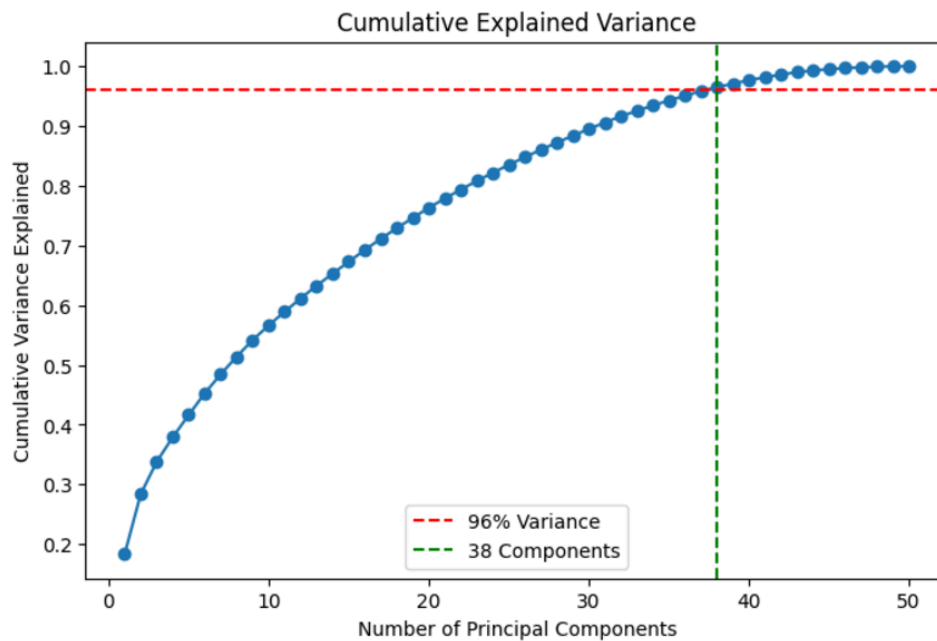
3.1 Principal Component Analysis (PCA)

While doing PCA, we got the relative importance of the principal components (left image) from which PC 1 explains 18.35% of the total variance, and PC 2 - 10%. Although PC 1 and PC 2 both explain more variance than any other component, when combined together, they explain only 28% of the whole variance. It might indicate that no small group of features dominates our dataset. Also, we found that 38 components explain 96% variance which is a lot of features, meaning that the dataset has high dimensionality with variability spread across many features. The plotted first 2 components are on the right image. We can see a clear clustering where points on the left belong to class 0, legitimate websites, and the points on the right to the origin are in class 1, phishing websites. This suggests that the data is separable between the first two principal components.

```
Principal Component 1: 0.1835
Principal Component 2: 0.1000
Principal Component 3: 0.0544
Principal Component 4: 0.0410
Principal Component 5: 0.0374
Principal Component 6: 0.0362
Principal Component 7: 0.0319
Principal Component 8: 0.0297
Principal Component 9: 0.0276
Principal Component 10: 0.0247
Principal Component 11: 0.0228
Principal Component 12: 0.0223
Principal Component 13: 0.0214
Principal Component 14: 0.0201
Principal Component 15: 0.0199
Principal Component 16: 0.0195
Principal Component 17: 0.0190
Principal Component 18: 0.0176
Principal Component 19: 0.0173
Principal Component 20: 0.0168
Principal Component 21: 0.0155
Principal Component 22: 0.0150
Principal Component 23: 0.0146
Principal Component 24: 0.0136
Principal Component 25: 0.0133
Principal Component 26: 0.0129
Principal Component 27: 0.0123
Principal Component 28: 0.0120
Principal Component 29: 0.0116
Principal Component 30: 0.0112
```



On the following graph, we can see that the first 10 components capture around 55% of the variance and combining with the next 10 components, they capture around 75%. We can see that the green and red lines represent getting 96% variability explained by the first 38 components. It also shows that the rest 12 components capture very little variance and are not critical for understanding the dataset.



3.2 Decision Trees

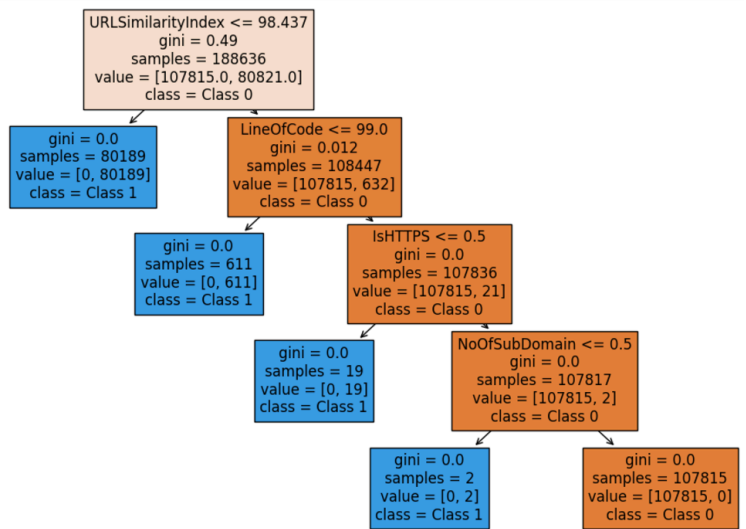
The first decision tree with a max depth of five got us 100% scores for all the evaluation metrics. From the feature importance, we have found that the URLSimilarityIndex has 98.6% importance. The results showed us that the accuracy is 100%, same as all the other metrics, which is suspiciously good. It could be explained by the results from the heatmap where labels had the highest correlation with URLSimilarityIndex, meaning that this feature was the most significant to our model.

Training Accuracy: 1.0

Testing Accuracy: 1.0

Classification Report:

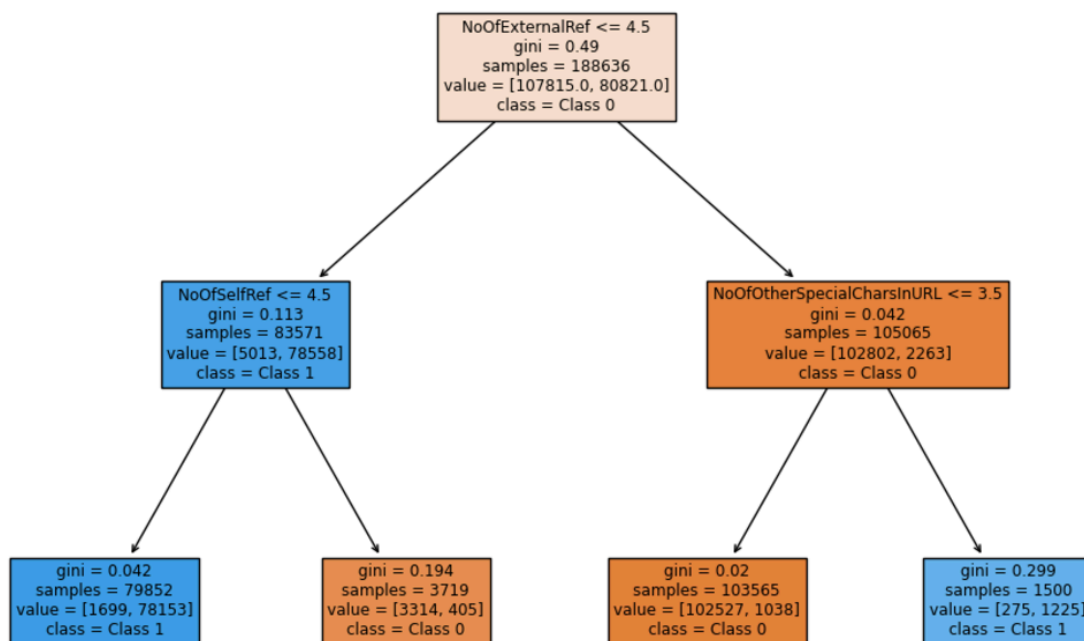
	precision	recall	f1-score	support
0	1.00	1.00	1.00	27035
1	1.00	1.00	1.00	20124
accuracy			1.00	47159
macro avg	1.00	1.00	1.00	47159
weighted avg	1.00	1.00	1.00	47159



However, the model could still overfit to the training data, even with a small max depth. So, we did the cross-validation to ensure it generalizes across different splits of the dataset, and got 100% accuracy once again. It means that the data does not overfit. That is why we looked into the initial research where it was found that the scholars collected the data from which the “URLSimilarityIndex” was calculated using a source URL and a target URL. Actually, their research includes “phishing URL detection framework based on similarity index”. So, the results are exceptional because scholars have come up with a formula that would make the prediction of phishing websites 100% precise. To see how the model would behave on the raw data, we have dropped some columns that were artificially calculated and used max depth of 2 to make a smaller tree. The results gave us 98% accuracy, indicating that the most significant feature is “NoOfExternalRef”.

Training Accuracy: 0.9818857482134905
 Testing Accuracy: 0.982293941771454
 Classification Report:

	precision	recall	f1-score	support
0	0.99	0.98	0.98	27035
1	0.98	0.98	0.98	20124
accuracy			0.98	47159
macro avg	0.98	0.98	0.98	47159
weighted avg	0.98	0.98	0.98	47159



To make sure the results are good throughout different splits, we calculated cross-validation accuracy once again and got a 98% score.

4. Experimental Method

In our experimental methods, we will use Random Forest, AdaBoost, and Multi-layer Perception (MLP) to compare the results. We will also exclude “URLSimilarityIndex”, “CharContinuationRate”, “URLTitleMatchScore”, “URLCharProb”, and “TLDDLegitimateProb” parameters from our scope as they were crafted by the scholars. In this project, the machine learning hypothesis is that certain features in the dataset are strong indicators of phishing and can be used to classify phishing websites effectively.

5. Results & Analysis

5.1 Random Forest

Applying a random forest algorithm gave us 99.98% accuracy scores for both testing and cross-validation. Compared to the last decision tree, here the feature importance is more spread out. Previously the most significant feature was “NoOfExternalRef” with 98%, while here its importance is equivalent to 15% only. The most important features here represent the line of code (how long it is), the number of hyperlinks, the number of images, and the amount of css used in the website. There is no need for further changes or modifications with such results.

```
Cross-Validation Accuracy (Random Forest): 0.9998515660351105
Testing Accuracy: 0.9998939756992302
Classification Report:
              precision    recall  f1-score   support

     0           1.00        1.00        1.00       27035
     1           1.00        1.00        1.00       20124

 accuracy                   1.00       47159
 macro avg           1.00        1.00        1.00       47159
 weighted avg        1.00        1.00        1.00       47159

Feature Importances:
              Feature  Importance
18      LineOfCode    0.159590
44      NoOfExternalRef 0.155216
42      NoOfSelfRef    0.133750
39      NoOfImage      0.111838
40      NoOfCSS        0.061259
31      HasSocialNet   0.058060
```

5.2 AdaBoost

Applying AdaBoost resulted in 99.93% testing and cross-validation accuracy scores, which is only 0.05% less than in the random forest model. However, the top-5 parameters changed a little with LineOfCode and NoOfExternalRef unchanged. New important parameters here are whether the website has “https”, the longest line of code, and the number of special characters in URL. The results are excellent, so no further changes are needed.

```

Training Accuracy: 0.9995228906465362
Testing Accuracy: 0.9993638541953815
Classification Report:
      precision    recall  f1-score   support

      0         1.00      1.00      1.00      27035
      1         1.00      1.00      1.00      20124

   accuracy          1.00          1.00      1.00      47159
  macro avg          1.00          1.00      1.00      47159
 weighted avg          1.00          1.00      1.00      47159

Cross-Validation Accuracy (AdaBoost): 0.9993320461744123
Feature Importances:
      Feature  Importance
17      IsHTTPS    0.160754
18      LineOfCode 0.123466
19      LargestLineLength 0.119853
44      NoOfExternalRef 0.109004
15      NoOfOtherSpecialCharsInURL 0.093276
4       NoOfSubDomain 0.079553
42      NoOfSelfRef 0.077023

```

5.3 Multi-layer Perceptron (MLP)

Before feeding the data into MLP, we scaled our features. The results are 99.97% testing and 99.98% cross-validation accuracy. Such ML performance suggests the dataset has features that can be used to perfectly differentiate phishing websites.

```

Training Accuracy: 0.9999522890646536
Testing Accuracy: 0.9997455416781527
Classification Report:
      precision    recall  f1-score   support

      0         1.00      1.00      1.00      27035
      1         1.00      1.00      1.00      20124

   accuracy          1.00          1.00      1.00      47159
  macro avg          1.00          1.00      1.00      47159
 weighted avg          1.00          1.00      1.00      47159

Cross-Validation Accuracy (MLP): 0.9998038547063247

```

5.4 Overall Summary

In this study, five features were removed from the dataset because they were derived through formulas or calculations rather than being part of the raw, original data. These derived potentially informative features were excluded to minimize the risk of overfitting or feature leakage. Among the features that remained, some had a moderate correlation with the target

variable and then played a significant role in the model's predictive performance. For example, “IsHTTPS” is a feature differentiating links with or without “https” that had -0.61 correlation with labels and, at the same time, was the most important feature in the AdaBoost model.

The results for all the experimental methods raise concerns in terms of the validity of the data. On the other hand, the dataset originates from a reliable source and prior research was made based on this specific data adds to its credibility. Moreover, no evidence of data leakage was found, and all the derived features crafted by the scholars were dropped to eliminate overfitting. Until no further investigation is done, it is hard to tell whether the results with the high accuracy scores imply that the dataset lacks reliability or that it is inherently easy to classify. Testing models on unseen real-world data would help to better understand the dataset and find out the underlying cause of this issue.

References.

- (1) Prasad, A. & Chandra, S. (2024). PhiUSIIL Phishing URL (Website) [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.1016/j.cose.2023.103545>.
- (2) <https://archive.ics.uci.edu/dataset/967/phiusiil+phishing+url+dataset>